BRIGHTTHREADS

A Data Analytics Case Study

Applying the OSEMN Framework to a 10% Sales-Growth Goal

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Introduction to Data Analytics, Capstone Activity: Obtain, Scrub, Explore, Model, iNterpret

Project Overview

This is my write-up for a course project that walks through the OSEMN framework (Obtain, Scrub, Explore, Model, iNterpret) from start to finish. The scenario follows Anna, who owns BrightThreads, a clothing boutique in New York with both a storefront and an online shop. Anna wants to grow total sales by 10% next year, and my job through this project was to use her data to figure out if that's realistic and what she should actually do about it.

Each section below covers one stage of the OSEMN process. I started by identifying and cleaning up the raw sales data, then explored and modeled it to find patterns worth acting on, and finished by tying it all back into the recommendations Anna presented to her team.

- Improve sales
- Grow our footprint
- Make customers happy

Target: grow total sales by 10% in the upcoming year.

Stage 1: Obtain & Scrub (Week 2)

Scenario: Anna is planning for next year and wants to grow total sales by 10%. Before any analysis can start, I need to figure out what data actually matters and get it into usable shape.

Framing the Analysis

Q: In this scenario, what is a SMART goal that would benefit from data analysis?

Grow BrightThreads' total sales, online and in-store combined, by 10% within the next 12 months. I'd measure this against current fiscal year revenue and check in monthly to make sure things are on pace.

Q: What is a Primary KPI that would be useful to analyze for this goal?

Total sales revenue, tracked month over month and year over year. I'd also keep an eye on average order value and units sold in the top categories as supporting metrics.

Q: What relevant data would you gather in this scenario, and how would you obtain it?

I'd want online and in-store transaction records (item, category, price, quantity, date), customer zip codes to spot any geographic patterns, social media ad spend and site-visit numbers, and some outside data on industry trends.

First-party data: sales exports from the online storefront and in-store POS system, plus a customer survey. Both come directly from BrightThreads' own systems.

Third-party data: trend forecasting data from an outside research firm, used to check BrightThreads' numbers against the wider apparel market.

For methods, that mostly means pulling direct data exports from the storefront and POS system, running a customer survey, and using web analytics to track site traffic and ad performance.

Data Validity

Q: Using what you know about data validity, do you think this data is valid? Why or why not?

Not yet. It's useful for getting a general sense of things, but going through the four segments turned up duplicate rows, inconsistent formatting, missing fields, and a few clear typos. This needs to be cleaned up before I'd trust it for any real decisions.

Q: What issue did you identify in Segment 1 of the data?

Duplicate rows. Customer 102537's order (#1025371) and customer 197528's order (#1975281) each show up twice with identical values. Left as-is, that would double-count revenue and order volume.

Q: What issue did you identify in Segment 2 of the data?

Inconsistent formatting. Some zip codes are plain 5-digit and others use ZIP+4 (98765-5842, 78459-0000), and one SKU (aB3C5dE7) breaks the all-caps pattern every other SKU follows. A couple of item_cost values ($70.00, $50.00) also don't match the $X9.99 pricing everything else uses, which makes me think they were rounded or typed in wrong.

Q: What issue did you identify in Segment 3 of the data?

Missing values. Customer 731952's row is missing item_sku, item_category, and item_cost, and customer 936125's row is missing order_number and customer_zip. Both need to be tracked down or dropped before I use this data.

Q: What issue did you identify in Segment 4 of the data?

A couple of data-entry mistakes. Customer 651927 has an item_cost of $0.069, which looks like a misplaced decimal point and should probably read $69.99. Customer 786415 shows $5,999.00, which is likely a missing

decimal and should read $59.99. On top of that, customer 567812 has a different zip code here (12345) than in Segment 2 (78459-0000), so that's a conflict that needs to be sorted out too.

Stage 2: Explore & Model (Week 3)

Scenario: With clean data in hand, Anna looks at last quarter's online sales to see what the numbers show and builds a few models to project what's coming next.

Exploring the Dataset

Q: What are some things you can tell about this dataset? What does its size tell you?

The Q1 online sales dataset has 58 order-level rows covering January through March. That's a small but workable sample. It's enough to spot patterns in price points and category mix, but I'd treat anything about seasonality as a rough sense of direction rather than a firm conclusion until there's more data to compare it against.

Q: What kind of data is in this dataset? (Numerical, categorical, etc.)

It's a mix. sale_date is temporal, customer_id, order_number, and zip_code are really categorical identifiers even though they're stored as numbers (they shouldn't be summed or averaged), item_category is categorical, and quant_items_per_order and order_total are numerical, discrete and continuous respectively.

Q: What is the minimum and maximum value in the order_total column?

Minimum is $39.99 and maximum is $149.99. The median order is $79.99, and most orders, roughly two-thirds, fall between $49.99 and $95.99.

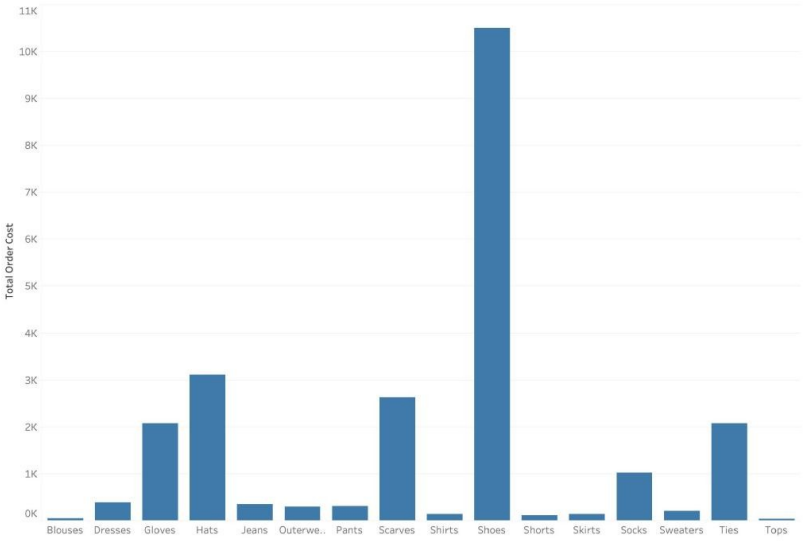
Q: What kind of chart would you use to help visualize this data?

A line chart to track order totals over time and check for trend or seasonality, a bar chart to compare total revenue across item categories, and a scatter plot to look at how order size and order total relate to each other.

Q: Would you add an additional column using feature engineering?

Yes. A day-of-week column pulled from sale_date would help show whether weekends or certain weekdays drive more orders. A month or week-number column would make seasonal patterns easier to spot, and an average-item-cost column (order_total divided by quant_items_per_order) would help separate growth from more items sold versus growth from pricier items.

Reading the Charts



Total order cost by item category, BrightThreads' current top sellers.

Q: What would be a good title for this chart?

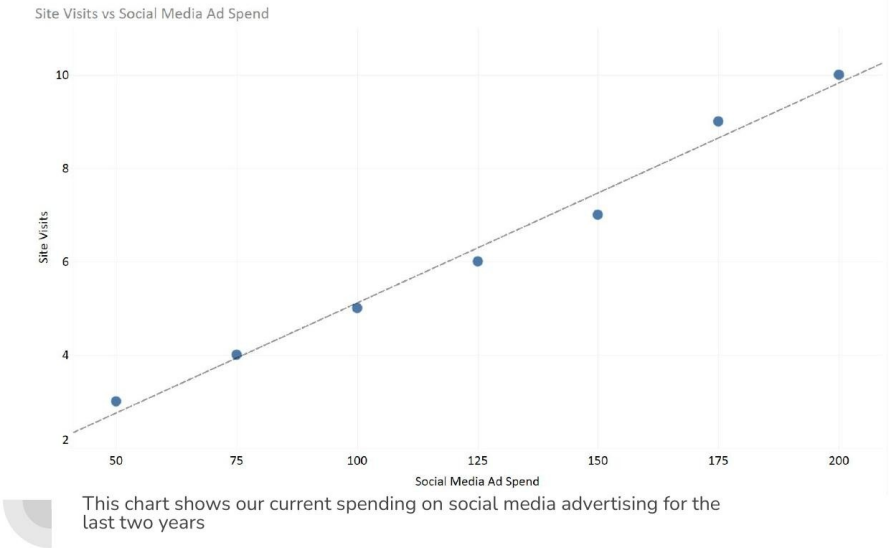
Something like "BrightThreads Revenue by Item Category." That gets across that it's comparing how much revenue each category brings in, not just how many units sell.

Q: What does this chart tell you about the number of orders relative to order amount?

Revenue is concentrated in a few categories instead of spread out evenly. Shoes alone bring in roughly three to four times what the next-highest category (scarves) does, while categories like blouses, shirts, and tops barely register. That points to either a lot more orders or higher per-item pricing in the top categories.

Q: What range do most of the orders tend to be in?

Most individual orders fall somewhere between $50 and $95, with the median sitting at $79.99. That's a fairly comfortable mid-range price point rather than a cluster at either extreme.



Site visits vs. social media ad spend.

Q: Do you notice any correlations between the variables in this chart? If so, how would you describe them?

Yes, there's a strong, almost perfect positive correlation (r is about 0.99) between social media ad spend and site visits. As weekly ad spend goes from $50 to $200, site visits climb in a pretty straight line from around 3 to 10. In this range, ad spend looks like a reliable, direct driver of traffic.

Modeling the Data

Q: Roughly how many site visits can be expected if the marketing budget is increased to $250?

I fit a simple linear regression to the weekly ad-spend and site-visit numbers (slope of about 0.047 visits per dollar, intercept around 0.35), which projects roughly 12 site visits per week at a $250 weekly spend. That's just continuing the trend that already holds up through $200.

Q: Is this model sufficient for general use? Why or why not?

It holds up well within the range of spend I actually have data for, $50 to $200, where the training and test lines stay close together. I'd be more careful extrapolating far past that, since ad spend often hits diminishing returns and the model hasn't actually been tested against numbers that high.

Q: A clustering model identified two customer groups. How would you describe them, and why?

One group looks like frequent, higher-value shoppers, customers with multiple orders and pricier items like outerwear or sweaters. The second group looks more like occasional or budget-conscious shoppers, buying single, lower-cost items like tops or blouses. This lines up with the gap I saw earlier between BrightThreads' high- and low-revenue categories.

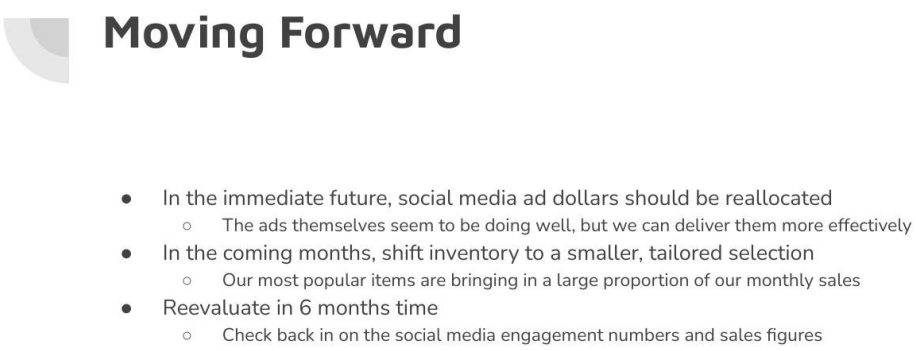
Q: What model would you use to forecast BrightThreads' sales for the coming quarter, and why?

I'd lean toward a time-series approach, something like seasonal decomposition or exponential smoothing, since the two-year sales chart shows a pattern that repeats every quarter with peaks around Q2 and Q4. A model built to account for that seasonality should forecast more accurately than a plain linear trend line.

Stage 3: iNterpret (Week 4)

Scenario: Anna presents her findings to BrightThreads stakeholders, turning the analysis into a clear story and a set of recommendations.

The Presentation



Closing slide, "Moving Forward," Anna's recommendations to stakeholders.

Q: What was the objective of this analysis?

To figure out whether BrightThreads can realistically grow total sales by 10% in the coming year, and to pin down the specific actions that would get them there.

Q: How does the data answer Anna's question?

It backs up the 10% goal. Historical sales, top-category revenue, and the strong link between ad spend and site visits all point to two concrete, low-risk moves (focusing inventory and reallocating ad spend) that are enough to close the gap without needing new investment.

Q: How can Anna apply this in a business context?

Anna can shift her existing social media ad budget toward the channels that are performing best instead of spending more overall, narrow her inventory toward proven top sellers like shoes, scarves, and hats, and check back on site-visit and sales numbers in six months to see if the plan is working.

What the question asks about	Where it appears in the deck
Recap of the project	Slides 1 to 2, BrightThreads Goals
Methods used	Slide 6, Methods For Analysis
Data visualizations	Slides 7 to 12: sales trend, target model, top sellers, prioritization model, ad-spend scatter, visits forecast
Explanation of the project	Slides 3 to 5, Project Objectives, Data We Analyzed
Recommendations	Slide 16, Moving Forward

Q: Which parts of the presentation were meant to explain, engage, and enlighten the audience?

Explain: the Project Objectives and Data We Analyzed slides walk the audience through what was studied and why. Engage: the six chart slides let stakeholders see the trends for themselves instead of just being told about them. Enlighten: the "What the Data Shows" and "Moving Forward" slides tie those visuals back to specific next steps people can act on.

Q: In your opinion, what parts of the presentation were the setup, buildup, climax, and conclusion? Why?

Setup: the opening goals slide and the question "Can we increase sales by 10% in the upcoming year?" It sets the stakes before any data shows up.

Buildup: the objectives, data sources, methods, and chart slides. Each one adds a bit more evidence and builds toward an answer.

Climax: the "So, can we increase sales by 10%? Absolutely!" slide. It's the payoff after all that buildup, delivered as a clear yes.

Conclusion: the "Moving Forward" slide. It turns the answer into three concrete actions stakeholders can act on right away.

Key Takeaways

Working through the full OSEMN cycle on BrightThreads' data pointed to a pretty clear, evidence-backed path to a 10% sales increase:

- Obtain and scrub: the raw sales data needed real cleanup. Duplicates, inconsistent formats, missing fields, and a few entry errors all had to be addressed before it could be trusted.
- Explore: revenue is concentrated in a handful of top categories, led by shoes, and most orders land in a fairly predictable $50 to $95 range.
- Model: social media ad spend is a strong, nearly linear predictor of site visits (r is about 0.99), and BrightThreads' sales follow a repeating quarterly pattern that's worth building a seasonal forecast around.
- iNterpret: shifting ad spend to the best-performing channels and narrowing inventory toward proven best sellers are two low-cost, high-confidence moves that make the 10% growth target realistic.

This project gave me hands-on practice with the full data analytics lifecycle, from raw, messy data through cleaning, exploration, predictive modeling, and telling the story to stakeholders at the end.